

Fractional integration associated with Zygmund dilations

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Abstract

First, we study a family of fractional integral operator defined as

$$\mathbf{I}_{\alpha\beta\vartheta} f(u, v, t) = \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \mathbf{V}^{\alpha\beta\vartheta}[(u, v, t) \odot (\xi, \eta, \tau)^{-1}] d\xi d\eta d\tau$$

where $\odot$ denotes the multiplication law on a real Heisenberg group.

$\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ satisfying Zygmund dilations. A characterization is established between $\mathbf{I}_{\alpha\beta\vartheta}: \mathbf{L}^p(\mathbb{R}^{2n+1}) \longrightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ and necessary constraints consisting of $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq 0$ for $1 < p < q < \infty$.

Moreover, denote $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$: I is an interval and $\mathbf{Q}, \mathbf{P}$ are cubes parallel to the coordinates. For $\alpha, \beta \in \mathbb{R}$, we consider

$$\mathbf{M}_{\alpha\beta} f(u, v, t) = \sup_{\mathbf{R} \ni (0,0,0): \text{vol}\{I\} = \text{vol}\{\mathbf{Q}\}^{\frac{1}{n}} \text{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \text{vol}\{\mathbf{Q}\}^{\frac{\alpha}{n}-1} \text{vol}\{\mathbf{P}\}^{\frac{\beta}{n}-1} \text{vol}\{I\}^{\beta-1} \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau.$$

We show $\mathbf{M}_{\alpha\beta}: \mathbf{L}^p(\mathbb{R}^{2n+1}) \longrightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ for $1 < p \leq q < \infty$ whenever $\frac{\alpha+\beta}{n+1} = \frac{1}{p} - \frac{1}{q}$.

1 Introduction

A fractional integral operator $\mathbf{T}_{\mathbf{a}}$ is initially defined on $\mathbb{R}^{\mathbf{N}}$ as

$$\mathbf{T}_{\mathbf{a}} f(x) = \int_{\mathbb{R}^{\mathbf{N}}} f(y) \left[\frac{1}{|x - y|} \right]^{\mathbf{N}-\mathbf{a}} dy, \quad 0 < \mathbf{a} < \mathbf{N}. \quad (1.1)$$

In 1928, Hardy and Littlewood [1] have obtained an regularity theorem for $\mathbf{T}_{\mathbf{a}}$ when $\mathbf{N} = 1$. Ten years later, Sobolev [2] made extensions on every higher dimensional space.

◇ Throughout, $\mathfrak{B} > 0$ is regarded as a generic constant depending on its sub-indices.

Hardy-Littlewood-Sobolev theorem *Let $\mathbf{T}_{\mathbf{a}}$ defined in (1.1) for $0 < \mathbf{a} < \mathbf{N}$. We have*

$$\begin{aligned} \|\mathbf{T}_{\mathbf{a}} f\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \quad 1 < p < q < \infty \\ \text{if and only if} \quad \frac{\mathbf{a}}{\mathbf{N}} &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1.2)$$

This classical result was first re-investigated by Folland and Stein [4] on Heisenberg groups.

In this paper, $\odot$ denotes a multiplication law:

$$(u, v, t) \odot (\xi, \eta, \tau) = \left[u + \xi, v + \eta, t + \tau + \mu(u \cdot \eta - v \cdot \xi) \right], \quad \mu \in \mathbb{R} \quad (1.3)$$

for every $(u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ and $(\xi, \eta, \tau)^{-1} = (-\xi, -\eta, -\tau) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$.

Let $0 < \delta < n + 1$. Consider

$$\mathbf{S}_\delta f(u, v, t) = \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \Omega^\delta \left[(u, v, t) \odot (\xi, \eta, \tau)^{-1} \right] d\xi d\eta d\tau. \quad (1.4)$$

Ω^δ is a distribution in $\mathbb{R}^{2n+1}$ agree with

$$\Omega^\delta(u, v, t) = \left[\frac{1}{|u|^2 + |v|^2 + |t|} \right]^{n+1-\delta}, \quad (u, v, t) \neq (0, 0, 0). \quad (1.5)$$

Folland-Stein theorem Let $\mathbf{S}_\delta$ defined in (1.4)-(1.5) for $0 < \delta < n + 1$. We have

$$\begin{aligned} \|\mathbf{S}_\delta f\|_{\mathbf{L}^q(\mathbb{R}^{n+1})} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p < q < \infty \\ \text{if and only if} \quad \frac{\delta}{n+1} &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1.6)$$

The best constant for the $\mathbf{L}^p \rightarrow \mathbf{L}^q$ -norm inequality in (1.6) is found by Frank and Lieb [13]. A discrete analogue of this result has been obtained by Pierce [14]. Recently, the regarding commutator estimates are established by Fanelli and Roncal [15].

Next, we give a multi-parameter extension to **Folland-Stein theorem** by replacing Ω^δ with a larger kernel having singularity on every coordinate subspace. From (1.5), we have

$$\Omega^\delta(u, v, t) \leq \left[\frac{1}{|u||v| + |t|} \right]^{n+1-\delta}, \quad (u, t) \neq (0, 0) \text{ or } (v, t) \neq (0, 0).$$

A direct computation shows

$$\begin{aligned} \left[\frac{1}{|u||v| + |t|} \right]^{n+1-\delta} &\approx \left[\frac{1}{|u|^2|v|^2 + t^2} \right]^{\frac{n+1}{2}-\frac{\delta}{2}} \\ &= |u|^{\frac{\delta}{2}-\frac{n+1}{2}} |v|^{\frac{\delta}{2}-\frac{n+1}{2}} |t|^{\frac{\delta}{2}-\frac{n+1}{2}} \left[\frac{|u||v||t|}{|u|^2|v|^2 + t^2} \right]^{\frac{n+1}{2}-\frac{\delta}{2}} \\ &= |u|^{\left[\frac{\delta}{2}+\frac{n-1}{2}\right]-n} |v|^{\left[\frac{\delta}{2}+\frac{n-1}{2}\right]-n} |t|^{\left[\frac{\delta}{2}-\frac{n-1}{2}\right]-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\left[\frac{n+1}{2}-\frac{\delta}{2}\right]} \end{aligned}$$

whenever $(u, t) \neq (0, 0)$ or $(v, t) \neq (0, 0)$. Above estimates lead us to the following assertion. Let $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq 0$. $\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ agree with

$$\mathbf{V}^{\alpha\beta\vartheta}(u, v, t) = |u|^{\alpha-n} |v|^{\alpha-n} |t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\vartheta}, \quad u \neq 0, v \neq 0, t \neq 0. \quad (1.7)$$

Remark 1.1. By taking into account $\alpha = \frac{\delta}{2} + \frac{n-1}{2}$, $\beta = \frac{\delta}{2} - \frac{n-1}{2}$ and $\vartheta = \frac{n+1}{2} - \frac{\delta}{2}$ for $0 < \delta < n+1$, we find $\alpha > n\beta$ and $\vartheta = \frac{n+1}{2} - \frac{\delta}{2} > \frac{\alpha-n\beta}{n+1}$.

Define

$$\mathbf{I}_{\alpha\beta\vartheta}f(u, v, t) = \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \mathbf{V}^{\alpha\beta\vartheta}[(u, v, t) \odot (\xi, \eta, \tau)^{-1}] d\xi d\eta d\tau. \quad (1. 8)$$

Observe that

$$\mathbf{V}^{\alpha\beta\vartheta}[(ru, sv, rst) \odot (r\xi, s\eta, r\tau)^{-1}] = r^{\alpha+\beta-n-1} s^{\alpha+\beta-n-1} \mathbf{V}^{\alpha\beta\vartheta}[(u, v, t) \odot (\xi, \eta, \tau)^{-1}], \quad r, s > 0. \quad (1. 9)$$

A convolution operator of this type is said to be associated with Zygmund dilations. Certain singular integral operators defined on a Heisenberg group associated with Zygmund dilations have been systematically studied, for example by Phong and Stein [5], Ricci and Stein [6] and Müller, Ricci and Stein [7]. Much less is known for fractional integrals in this direction.

Theorem One Let $\mathbf{I}_{\alpha\beta\vartheta}$ defined in (1. 7)-(1. 8) for $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq 0$. We have

$$\begin{aligned} \|\mathbf{I}_{\alpha\beta\vartheta}f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} &\leq \mathfrak{B}_{p\ q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p < q < \infty \\ \text{if and only if} \quad \vartheta &\geq \frac{|\alpha - n\beta|}{n+1}, \quad \frac{\alpha + \beta}{n+1} = \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1. 10)$$

Denote $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$: I is an interval and $\mathbf{Q}, \mathbf{P}$ are cubes parallel to the coordinates. For $\alpha, \beta \in \mathbb{R}$, we define

$$\begin{aligned} \mathbf{M}_{\alpha\beta}f(u, v, t) = \\ \sup_{\mathbf{R} \ni (0,0,0): \text{vol}\{I\} = \text{vol}\{\mathbf{Q}\}^{\frac{1}{n}} \text{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \text{vol}\{\mathbf{Q}\}^{\frac{\alpha}{n}-1} \text{vol}\{\mathbf{P}\}^{\frac{\beta}{n}-1} \text{vol}\{I\}^{\beta-1} \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau. \end{aligned} \quad (1. 11)$$

For $\mathbf{M}_{\alpha\beta}$ defined in the Euclidean space, in particular at $\alpha = \beta = 0$, the regarding $\mathbf{L}^p$ -theorem and its weighted analogue have been well established. See the papers by Ricci and Stein [6] and Fefferman and Pipher [10].

Theorem Two Let $\mathbf{M}_{\alpha\beta}$ defined in (1. 11) for $\alpha, \beta \in \mathbb{R}$. We have

$$\begin{aligned} \|\mathbf{M}_{\alpha\beta}f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} &\leq \mathfrak{B}_{p\ q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p \leq q < \infty \\ \text{whenever} \quad \frac{\alpha + \beta}{n+1} &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1. 12)$$

Later, we shall find

$$\mathbf{M}_{\alpha\beta}f(u, v, t) \leq \mathbf{M}_{\gamma}f(u, v, t), \quad \gamma = \frac{\alpha+\beta}{n+1} \quad (1. 13)$$

for which $\mathbf{M}_{\gamma}, 0 \leq \gamma < 1$ is defined as

$$\mathbf{M}_{\gamma}f(u, v, t) = \sup_{\mathbf{R} \ni (0,0,0)} \text{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau. \quad (1. 14)$$

Remark 1.2. The supremum in (1. 14) is taking over any rectangle $\mathbf{R} \subset \mathbb{R}^{2n+1}$ parallel to the coordinates.

$\mathbf{M}_0$ in (1. 14) is known as the strong maximal operator defined on Heisenberg groups. The L^p -boundedness of $\mathbf{M}_0$ has been proved by M. Christ [11] using a mixture of techniques developed previously by Ricci and Stein [8]-[9] and Christ [12] for singular integrals defined on sub-manifolds within a general setting of Nilpotent Lie groups. **Theorem Two** will be a corollary of the next result.

Theorem Three Let $\mathbf{M}_\gamma$ defined in (1. 14) for $0 \leq \gamma < 1$. We have

$$\begin{aligned} \|\mathbf{M}_\gamma f\|_{L^q(\mathbb{R}^{2n+1})} &\leq \mathfrak{B}_{p,q} \|f\|_{L^p(\mathbb{R}^{2n+1})}, \quad 1 < p \leq q < \infty \\ \text{whenever} \quad \gamma &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \tag{1. 15}$$

The remaining paper is organized as follows. In section 2, we show that two constraints consisting of $\alpha, \beta, \vartheta, p, q$ in (1. 10) are necessary conditions. In section 3, we prove their sufficiency and finish the proof of **Theorem One**. In section 4, we prove **Theorem Three** which implies **Theorem Two** with a new approach by applying a geometric covering lemma due to Córdoba and Fefferman [3].

2 Some necessary constraints

Let $\mathbf{I}_{\alpha\beta\vartheta}$ defined in (1. 7)-(1. 8). By changing variable $\tau \longrightarrow \tau - \mu(u \cdot \eta - v \cdot \xi)$, we find

$$\begin{aligned} \mathbf{I}_{\alpha\beta\vartheta} f(u, v, t) &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \mathbf{V}^{\alpha\beta\vartheta} [u - \xi, v - \eta, t - \tau - \mu(u \cdot \eta - v \cdot \xi)] d\xi d\eta d\tau \\ &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \mathbf{V}^{\alpha\beta\vartheta}(u - \xi, v - \eta, t - \tau) d\xi d\eta d\tau \\ &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \\ &\quad |u - \xi|^{\alpha-n} |v - \eta|^{\alpha-n} |t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau. \end{aligned} \tag{2. 1}$$

Consider a more general situation by replacing $\mathbf{V}^{\alpha\beta\vartheta}(u, v, t)$ with

$$|u|^{\alpha_1-n} |v|^{\alpha_2-n} |t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\vartheta}, \quad \alpha_1, \alpha_2, \beta \in \mathbb{R}, \quad \vartheta \geq 0 \tag{2. 2}$$

for $u \neq 0, v \neq 0, t \neq 0$.

Let $(u, v, t) \longrightarrow (ru, sv, rs\lambda t)$ and $(\xi, \eta, \tau) \longrightarrow (r\xi, s\eta, rs\lambda\tau)$ for $r, s > 0$ and $0 < \lambda < 1$ or $\lambda > 1$.

We have

$$\begin{aligned}
& \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iiint_{\mathbb{R}^{2n+1}} f \left[r^{-1} \xi, s^{-1} \eta, r^{-1} s^{-1} \lambda^{-1} [\tau - \mu \lambda (u \cdot \eta - v \cdot \xi)] \right] \right. \right. \\
& \quad \left. \left. |u - \xi|^{\alpha_1 - n} |v - \eta|^{\alpha_2 - n} |t - \tau|^{\beta - 1} \left[\frac{|u - \xi| |v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi| |v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \right\}^q dudvdt \right\}^{\frac{1}{q}} \\
&= r^{\alpha_1 + \beta} s^{\alpha_2 + \beta} r^{\frac{n+1}{q}} s^{\frac{n+1}{q}} \lambda^{\beta} \lambda^{\frac{1}{q}} \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \right. \right. \\
& \quad \left. \left. |u - \xi|^{\alpha_1 - n} |v - \eta|^{\alpha_2 - n} |t - \tau|^{\beta - 1} \left[\frac{|u - \xi| |v - \eta|}{\lambda |t - \tau|} + \frac{\lambda |t - \tau|}{|u - \xi| |v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \right\}^q dudvdt \right\}^{\frac{1}{q}} \quad (2.3) \\
&\geq r^{\alpha_1 + \beta} s^{\alpha_2 + \beta} r^{\frac{n+1}{q}} s^{\frac{n+1}{q}} \lambda^{\beta} \lambda^{\frac{1}{q}} \begin{cases} \lambda^{\vartheta}, & 0 < \lambda < 1, \\ \lambda^{-\vartheta}, & \lambda > 1 \end{cases} \\
& \quad \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \right. \right. \\
& \quad \left. \left. |u - \xi|^{\alpha_1 - n} |v - \eta|^{\alpha_2 - n} |t - \tau|^{\beta - 1} \left[\frac{|u - \xi| |v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi| |v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \right\}^q dudvdt \right\}^{\frac{1}{q}}.
\end{aligned}$$

The $\mathbf{L}^p \longrightarrow \mathbf{L}^q$ -norm inequality in (1. 15) implies that the last line of (2. 3) is bounded by

$$\left\{ \iiint_{\mathbb{R}^{2n+1}} \left| f(r^{-1} \xi, s^{-1} \eta, r^{-1} s^{-1} \lambda^{-1} \tau) \right|^p d\xi d\eta d\tau \right\}^{\frac{1}{p}} = r^{\frac{n+1}{p}} s^{\frac{n+1}{p}} \lambda^{\frac{1}{p}} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \quad (2.4)$$

This must be true for every $r, s > 0$ and $0 < \lambda < 1$ or $\lambda > 1$. We necessarily have

$$\frac{\alpha_1 + \beta}{n+1} = \frac{1}{p} - \frac{1}{q} = \frac{\alpha_2 + \beta}{n+1}, \quad \beta + \vartheta \geq \frac{1}{p} - \frac{1}{q} \quad \text{or} \quad \beta - \vartheta \leq \frac{1}{p} - \frac{1}{q}. \quad (2.5)$$

The first constraint in (2. 5) forces us to have $\alpha_1 = \alpha_2$. Therefore, write

$$\frac{\alpha + \beta}{n+1} = \frac{1}{p} - \frac{1}{q}, \quad \alpha = \alpha_1 = \alpha_2. \quad (2.6)$$

By bringing (2. 6) to the two inequalities in (2. 5), we find

$$\vartheta \geq \beta - \frac{\alpha + \beta}{n+1} = \frac{n\beta - \alpha}{n+1} \quad \text{or} \quad \vartheta \geq \frac{\alpha + \beta}{n+1} - \beta = \frac{\alpha - n\beta}{n+1}. \quad (2.7)$$

Together, we conclude $\vartheta \geq \frac{|\alpha - n\beta|}{n+1}$.

3 Proof of Theorem One

Given $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq \frac{|\alpha - n\beta|}{n+1}$, $\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ agree with $\mathbf{V}^{\alpha\beta\vartheta}(u, v, t)$ in (1. 7) whenever $u \neq 0, v \neq 0, t \neq 0$.

Suppose $\alpha \geq n\beta$. We have $\frac{|\alpha-n\beta|}{n+1} = \frac{\alpha-n\beta}{n+1}$ and

$$\begin{aligned}
\mathbf{V}^{\alpha\beta\vartheta}(\xi, \eta, \tau) &\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\frac{\alpha-n\beta}{n+1}} \\
&\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|u||v|}{|t|} \right]^{-\frac{\alpha-n\beta}{n+1}} \\
&= |u|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t|^{\frac{\alpha+\beta}{n+1}-1}, \quad u \neq 0, v \neq 0, t \neq 0.
\end{aligned} \tag{3. 1}$$

Suppose $\alpha \leq n\beta$. We find $\frac{|\alpha-n\beta|}{n+1} = \frac{n\beta-\alpha}{n+1}$ and

$$\begin{aligned}
\mathbf{V}^{\alpha\beta\vartheta}(u, v, t) &\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{\frac{\alpha-n\beta}{n+1}} \\
&\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|t|}{|u||v|} \right]^{\frac{\alpha-n\beta}{n+1}} \\
&= |u|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t|^{\frac{\alpha+\beta}{n+1}-1}, \quad u \neq 0, v \neq 0, t \neq 0.
\end{aligned} \tag{3. 2}$$

Let $\mathbf{I}_{\alpha\beta\vartheta}f$ defined in (1. 7)-(1. 8) and

$$\frac{\alpha + \beta}{n + 1} = \frac{1}{p} - \frac{1}{q}, \quad 1 < p < q < \infty. \tag{3. 3}$$

By changing variable $\tau \longrightarrow \tau + \mu(u \cdot \eta - v \cdot \xi)$, we have

$$\begin{aligned}
\mathbf{I}_{\alpha\beta\vartheta}f(u, v, t) &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \\
&\quad |u - \xi|^{\alpha-n}|v - \eta|^{\alpha-n}|t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \\
&\leq \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \\
&\quad |u - \xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t - \tau|^{\frac{\alpha+\beta}{n+1}-1} d\xi d\eta d\tau \quad \text{by (3. 1)-(3. 2).}
\end{aligned} \tag{3. 4}$$

Because $\mathbf{V}^{\alpha\beta\vartheta}$ is positive definite, it is suffice to assert $f \geq 0$. Define

$$\mathbf{F}_{\alpha\beta}(\xi, \eta, u, v, t) = \int_{\mathbb{R}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) |t - \tau|^{\frac{\alpha+\beta}{n+1}-1} d\tau. \tag{3. 5}$$

From (3. 4)-(3. 5), we find

$$\mathbf{I}_{\alpha\beta\vartheta}f(u, v, t) \leq \iint_{\mathbb{R}^{2n}} |u - \xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} \mathbf{F}_{\alpha\beta}(\xi, \eta, u, v, t) d\xi d\eta. \tag{3. 6}$$

Recall the **Hardy-Littlewood-Sobolev theorem** stated in the beginning of this paper. By applying (1. 2) with $\mathbf{a} = \frac{\alpha+\beta}{n+1}$ and $\mathbf{N} = 1$, we have

$$\begin{aligned} \left\{ \int_{\mathbb{R}} \mathbf{F}_{\alpha\beta}^q(\xi, \eta, u, v, t) dt \right\}^{\frac{1}{q}} &\leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}} [f(\xi, \eta, t - \mu(u \cdot \eta - v \cdot \xi))]^p dt \right\}^{\frac{1}{p}} \\ &= \mathfrak{B}_{p,q} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} \end{aligned} \quad (3. 7)$$

regardless of $(u, v) \in \mathbb{R}^n \times \mathbb{R}^n$.

On the other hand, by applying (1. 2) with $\mathbf{a} = n[\frac{\alpha+\beta}{n+1}]$ and $\mathbf{N} = n$, we find

$$\left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} d\xi \right\}^q du \right\}^{\frac{1}{q}} \leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}^n} \|f(u, \eta, \cdot)\|_{L^p(\mathbb{R})}^p du \right\}^{\frac{1}{p}} \quad (3. 8)$$

and

$$\left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} d\eta \right\}^q dv \right\}^{\frac{1}{q}} \leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}^n} \|f(\xi, v, \cdot)\|_{L^p(\mathbb{R})}^p dv \right\}^{\frac{1}{p}}. \quad (3. 9)$$

From (3. 6), we have

$$\begin{aligned} &\|\mathbf{I}_{\alpha\beta\vartheta} f\|_{L^q(\mathbb{R}^{2n+1})} \\ &\leq \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iint_{\mathbb{R}^{2n}} |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \mathbf{F}_{\alpha\beta}^q(\xi, \eta, u, v, t) d\xi d\eta \right\}^q dudvdt \right\}^{\frac{1}{q}} \\ &\leq \left\{ \iint_{\mathbb{R}^{2n}} \left\{ \iint_{\mathbb{R}^{2n}} |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \left\{ \int_{\mathbb{R}} \mathbf{F}_{\alpha\beta}^q(\xi, \eta, u, v, t) dt \right\}^{\frac{1}{q}} d\xi d\eta \right\}^q dudv \right\}^{\frac{1}{q}} \\ &\quad \text{by Minkowski integral inequality} \\ &\leq \mathfrak{B}_{p,q} \left\{ \iint_{\mathbb{R}^{2n}} \left\{ \iint_{\mathbb{R}^{2n}} |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} d\xi d\eta \right\}^q dudv \right\}^{\frac{1}{q}} \quad \text{by (3. 7)} \\ &\leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(u, \eta, \cdot)\|_{L^p(\mathbb{R})} d\eta \right\}^p du \right\}^{\frac{q}{p}} dv \right\}^{\frac{1}{q}} \quad \text{by (3. 8)} \\ &\leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(u, \eta, \cdot)\|_{L^p(\mathbb{R})} d\eta \right\}^q dv \right\}^{\frac{p}{q}} du \right\}^{\frac{1}{p}} \\ &\quad \text{by Minkowski integral inequality} \\ &\leq \mathfrak{B}_{p,q} \left\{ \iint_{\mathbb{R}^{2n}} \|f(u, v, \cdot)\|_{L^p(\mathbb{R})}^p dudv \right\}^{\frac{1}{p}} \quad \text{by (3. 9)} \\ &= \mathfrak{B}_{p,q} \|f\|_{L^p(\mathbb{R}^{2n+1})}. \end{aligned} \quad (3. 10)$$

4 Proof of Theorem Three

Recall $\mathbf{M}_\gamma$ defined in (1. 14) for $0 \leq \gamma < 1$. By taking $\xi \longrightarrow u - \xi$, $\eta \longrightarrow v - \eta$ and $\tau \longrightarrow t - \tau$, $\mathbf{M}_\gamma$ can be equivalently defined as

$$\mathbf{M}_\gamma f(u, v, t) = \sup_{\mathbf{R} \ni (u, v, t)} \mathbf{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau. \quad (4. 11)$$

Similarly, $\mathbf{M}_{\alpha\beta}$ defined in (1. 11) is equivalent to

$$\begin{aligned} \mathbf{M}_{\alpha\beta} f(u, v, t) = & \sup_{\mathbf{R} \ni (u, v, t): \mathbf{vol}\{I\} = \mathbf{vol}\{Q\}^{\frac{1}{n}} \mathbf{vol}\{P\}^{\frac{1}{n}}} \\ & \mathbf{vol}\{Q\}^{\frac{\alpha}{n}-1} \mathbf{vol}\{P\}^{\frac{\beta}{n}-1} \mathbf{vol}\{I\}^{\beta-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \end{aligned} \quad (4. 12)$$

where $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times \mathbf{I} \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$.

Note that $\mathbf{vol}\{I\} = \mathbf{vol}\{Q\}^{\frac{1}{n}} \mathbf{vol}\{P\}^{\frac{1}{n}}$ implies $\mathbf{vol}\{\mathbf{R}\} = \mathbf{vol}\{Q\}^{1+\frac{1}{n}} \mathbf{vol}\{P\}^{1+\frac{1}{n}}$. From (4. 12), we find

$$\begin{aligned} \mathbf{M}_{\alpha\beta} f(u, v, t) = & \sup_{\mathbf{R} \ni (u, v, t): \mathbf{vol}\{I\} = \mathbf{vol}\{Q\}^{\frac{1}{n}} \mathbf{vol}\{P\}^{\frac{1}{n}}} \\ & \mathbf{vol}\{Q\}^{\left[\frac{\alpha+\beta}{n+1}-1\right]\left(1+\frac{1}{n}\right)} \mathbf{vol}\{P\}^{\left[\frac{\alpha+\beta}{n+1}-1\right]\left(1+\frac{1}{n}\right)} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \\ = & \sup_{\mathbf{R} \ni (u, v, t): \mathbf{vol}\{I\} = \mathbf{vol}\{Q\}^{\frac{1}{n}} \mathbf{vol}\{P\}^{\frac{1}{n}}} \mathbf{vol}\{\mathbf{R}\}^{\frac{\alpha+\beta}{n+1}-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \quad (4. 13) \\ \leq & \sup_{\mathbf{R} \ni (u, v, t)} \mathbf{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \quad (\gamma = \frac{\alpha+\beta}{n+1}) \\ = & \mathbf{M}_\gamma f(u, v, t). \end{aligned}$$

Hence, $\mathbf{M}_{\alpha\beta}$ is controlled by the strong fractional maximal operator $\mathbf{M}_\gamma$ whenever $\gamma = \frac{\alpha+\beta}{n+1}$. Let $\gamma = \frac{1}{p} - \frac{1}{q}$, $1 < p \leq q < \infty$. In order to prove the converse, we need the following multi-parameter covering lemma.

Córdoba-Fefferman covering lemma

Let $\{\mathbf{R}_j\}_{j=1}$ be a collection of rectangles in $\mathbb{R}^{2n+1}$ parallel to the coordinates. There is a subsequence $\{\widehat{\mathbf{R}}_k\}_{k=1}$ such that

$$\mathbf{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \lesssim \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \quad (4. 14)$$

and

$$\left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{L^p(\mathbb{R}^{2n+1})}^p \leq \mathfrak{B}_p \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}, \quad 1 < p < \infty \quad (4. 15)$$

where χ is an indicator function.

Remark 4.1. This covering lemma is established by Córdoba and Fefferman [3] within a much more general setting. Namely, the Lebesgue measure is replaced by an absolutely continuous measure satisfying the rectangle A_∞ -property. In the following subsection, we prove this covering lemma w.r.t Lebesgue measure more directly.

Define

$$\mathbf{U}_\lambda = \left\{ (u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_\gamma f(u, v, t) > \lambda \right\}. \quad (4.16)$$

Given any $(u, v, t) \in \mathbf{U}_\lambda$, there is a rectangle $\mathbf{R}_j \ni (u, v, t)$ such that

$$\mathbf{vol}\{\mathbf{R}_j\}^{\gamma-1} \iiint_{\mathbf{R}_j} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau > \frac{1}{2} \lambda. \quad (4.17)$$

Let (u, v, t) run through the set $\mathbf{U}_\lambda$. We have

$$\mathbf{U}_\lambda \subset \bigcup_j \mathbf{R}_j. \quad (4.18)$$

By applying the covering lemma, we select a subsequence $\{\widehat{\mathbf{R}}_k\}_{k=1}^\infty$ from the union above and

$$\mathbf{vol}\{\mathbf{U}_\lambda\} \lesssim \mathbf{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \lesssim \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \quad \text{by (4.14)}. \quad (4.19)$$

Furthermore, we have

$$\begin{aligned} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} &\leq \sum_k \mathbf{vol}\{\widehat{\mathbf{R}}_k\} \\ &\lesssim \lambda^{-\frac{1}{1-\gamma}} \sum_k \left\{ \iiint_{\widehat{\mathbf{R}}_k} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \right\}^{\frac{1}{1-\gamma}} \quad \text{by (4.17)} \\ &\lesssim \lambda^{-\frac{1}{1-\gamma}} \left\{ \sum_k \iiint_{\widehat{\mathbf{R}}_k} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \right\}^{\frac{1}{1-\gamma}} \quad (0 \leq \gamma < 1) \\ &= \lambda^{-\frac{1}{1-\gamma}} \left\{ \iiint_{\mathbb{R}^{2n+1}} \left| f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi)) \sum_k \chi_{\widehat{\mathbf{R}}_k}(\xi, \eta, \tau) \right| d\xi d\eta d\tau \right\}^{\frac{1}{1-\gamma}} \quad (4.20) \\ &\leq \lambda^{-\frac{1}{1-\gamma}} \left\{ \iiint_{\mathbb{R}^{2n+1}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))|^p d\xi d\eta d\tau \right\}^{\frac{1}{p} \frac{1}{1-\gamma}} \left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}^{\frac{1}{1-\gamma}} \\ &\quad \text{by Hölder inequality} \\ &= \lambda^{-\frac{1}{1-\gamma}} \left\{ \iint_{\mathbb{R}^{2n}} \|f(\xi, \eta, \cdot)\|_{\mathbf{L}^p(\mathbb{R})}^p d\xi d\eta \right\}^{\frac{1}{p} \frac{1}{1-\gamma}} \left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}^{\frac{1}{1-\gamma}} \\ &\leq \lambda^{-\frac{1}{1-\gamma}} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}^{\frac{1}{1-\gamma}} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{p-1}{p} \frac{1}{1-\gamma}} \quad \text{by (4.15)}. \end{aligned}$$

By raising both sides of (4. 20) to the $(1 - \gamma)$ -th power and then taking into account for $1 - \gamma - \frac{p-1}{p} = \frac{1}{p} - \left[\frac{1}{p} - \frac{1}{q}\right] = \frac{1}{q}$, we find

$$\mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{q}} \lesssim \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \quad (4. 21)$$

Let $\mathbf{U}_\lambda$ defined in (4. 16). From (4. 19) and (4. 21), we obtain

$$\begin{aligned} \mathbf{vol}\left\{(u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_\gamma f(u, v, t) > \lambda\right\}^{\frac{1}{q}} &\lesssim \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{q}} \\ &\lesssim \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \end{aligned} \quad (4. 22)$$

By using this weak type (p, q) -estimate and applying Marcinkiewicz interpolation theorem, we conclude that $\mathbf{M}_\gamma$ is bounded from $\mathbf{L}^p(\mathbb{R}^{2n+1})$ to $\mathbf{L}^q(\mathbb{R}^{2n+1})$ if $\gamma = \frac{1}{p} - \frac{1}{q}$, $1 < p \leq q < \infty$.

4.1 A simpler proof of the covering lemma w.r.t Lebesgue measure

We select $\widehat{\mathbf{R}}_k$ from $\{\mathbf{R}_j\}_{j=1}$ as follows. Let $\widehat{\mathbf{R}}_1 = \mathbf{R}_1$. Having chosen $\widehat{\mathbf{R}}_1, \widehat{\mathbf{R}}_2, \dots, \widehat{\mathbf{R}}_{k-1}$, we pick $\widehat{\mathbf{R}}_k$ as the first rectangle $\mathbf{R}$ on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_{k-1}$ so that

$$\mathbf{vol}\left\{\mathbf{R} \cap \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell\right\} \leq \frac{1}{2} \mathbf{vol}\{\mathbf{R}\}. \quad (4. 23)$$

Suppose that $\mathbf{R}$ is an unselected rectangle. There is a positive number N such that $\mathbf{R}$ is on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_N$ and

$$\mathbf{vol}\left\{\mathbf{R} \cap \bigcup_{k=1}^N \widehat{\mathbf{R}}_k\right\} > \frac{1}{2} \mathbf{vol}\{\mathbf{R}\}. \quad (4. 24)$$

Let $\mathbf{M}$ be the strong maximal operator defined in $\mathbb{R}^{2n+1}$ which is $\mathbf{L}^p$ -bounded for $1 < p < \infty$. From (4. 24), we find

$$\mathbf{M}\chi_{\bigcup_k \widehat{\mathbf{R}}_k}(u, v, t) > \frac{1}{2}, \quad (u, v, t) \in \bigcup_j \mathbf{R}_j. \quad (4. 25)$$

By applying the $\mathbf{L}^2$ -boundedness of $\mathbf{M}$, we have

$$\begin{aligned} \mathbf{vol}\left\{\bigcup_j \mathbf{R}_j\right\} &= \iiint_{\bigcup_j \mathbf{R}_j} dudvdt \\ &\leq 2^2 \iiint_{\bigcup_j \mathbf{R}_j} [\mathbf{M}\chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \quad \text{by (4. 25)} \\ &\lesssim \iiint_{\mathbb{R}^{2n+1}} [\chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \\ &= \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} dudvdt = \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}. \end{aligned} \quad (4. 26)$$

On the other hand, (4. 23) suggests

$$\mathbf{vol} \left\{ \widehat{\mathbf{R}}_k \cap \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell \right\} \leq \frac{1}{2} \mathbf{vol} \{ \widehat{\mathbf{R}}_k \}, \quad k = 1, 2, \dots \quad (4. 27)$$

which further implies

$$\mathbf{vol} \left\{ \widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell \right\} > \frac{1}{2} \mathbf{vol} \{ \widehat{\mathbf{R}}_k \}, \quad k = 1, 2, \dots \quad (4. 28)$$

Let $\phi \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})$ and $\|\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} = 1$. We have

$$\begin{aligned} & \iiint_{\mathbb{R}^{2n+1}} \phi(\xi, \eta, \tau) \sum_k \chi_{\widehat{\mathbf{R}}_k}(\xi, \eta, \tau) d\xi d\eta d\tau \\ &= \sum_k \iiint_{\widehat{\mathbf{R}}_k} \phi(\xi, \eta, \tau) d\xi d\eta d\tau \\ &= \sum_k \left\{ \mathbf{vol} \{ \widehat{\mathbf{R}}_k \}^{-1} \iiint_{\widehat{\mathbf{R}}_k} \phi(\xi, \eta, \tau) d\xi d\eta d\tau \right\} \mathbf{vol} \{ \widehat{\mathbf{R}}_k \} \\ &< 2 \sum_k \left\{ \mathbf{vol} \{ \widehat{\mathbf{R}}_k \}^{-1} \iiint_{\widehat{\mathbf{R}}_k} |\phi(\xi, \eta, \tau)| d\xi d\eta d\tau \right\} \mathbf{vol} \left\{ \widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell \right\} \quad \text{by (4. 28)} \quad (4. 29) \\ &\lesssim \sum_k \int_{\widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell} \left\{ \mathbf{vol} \{ \widehat{\mathbf{R}}_k \}^{-1} \iiint_{\widehat{\mathbf{R}}_k} |\phi(\xi, \eta, \tau)| d\xi d\eta d\tau \right\} dudvdt \\ &\leq \sum_k \iiint_{\widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell} \mathbf{M}\phi(u, v, t) dudvdt \\ &= \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} \mathbf{M}\phi(u, v, t) dudvdt. \end{aligned}$$

Hölder inequality further implies

$$\begin{aligned} \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} \mathbf{M}\phi(u, v, t) dudvdt &\leq \|\mathbf{M}\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} \mathbf{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}^{\frac{1}{p}} \\ &\leq \mathfrak{B}_p \|\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} \mathbf{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}^{\frac{1}{p}} \quad \mathbf{M} \text{ bounded on } \mathbf{L}^p(\mathbb{R}^{2n+1}) \\ &= \mathfrak{B}_p \mathbf{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}^{\frac{1}{p}}. \end{aligned} \quad (4. 30)$$

By substituting (4. 30) to (4. 29) and taking the supremum of ϕ , we arrive at

$$\left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \mathbf{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}^{\frac{1}{p}}, \quad 1 < p < \infty. \quad (4. 31)$$

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